Proficient in AutoCAD (Mechanical) (80 Hours)

S.No	Session	Topics	Type	Duration (hrs)
1	Session 1&2	**AN INTRODUCTION TO ENGINEERING DRAWINGS AND AUTOCAD** Introduction to Engineering Drawings Applications of AutoCAD. (Machine drawings, Production drawings, Part drawings, Assembly drawings) Introduction to AutoCAD and AutoCAD Mechanical Exploring GUI and Workspaces Co-ordinate systems **Display Control:** Zoom, Pan, Redraw, Regen, Clean Screen, Steering wheels File Management - New, Qnew, Open, Save, Save as, Close, Exit, Quit	Theory/Demo	4 HRS
2	Session 3&4	Exercises Included for practice	Practicals	4 HRS
3	Session 5&6	**DRAFTING BASIC GEOMETRY SHAPES IN AUTOCAD** **Drawing settings:** Units, Limits **Drafting settings:** Snap and Grid, Polar tracking, Object snap, 3D Object snap, Dynamic input (F1 to F12 Function keys) **Drawing tools:** Line, Circle, Arc, Ellipse, Donut, Polygon, Rectangle, Point, Multiple point **Modify tools:** Erase, Undo, Redo, Explode, Move, Copy, Rotate, Mirror, Array, Align, Scale, Stretch, Lengthen, Trim, Extend, Break, Join, Chamfer, Fillet, Blend curves, Object selection **Object properties:** Properties window, Color, Linetype, Linetype scale, Line weight, Match properties	Theory/Demo	4 HRS
4	Session 7&8	Exercises Included for practice	Practicals	4 HRS
5	Session 9&10	**ADVANCED DRAFTING AND MODIFYING TOOLS IN AUTOCAD** **Drawing tools:** Multiline, Poly line, Spline (Fit & CV), Construction Line, Ray, Wipeout, Revision cloud, Helix, Region **Modify tools:** Multiline style, Multiline edit, Polyline edit, Spline edit, Offset, PD Mode, Grip editing, PD Size, Edit Array, Change space, Reverse, Copy nested objects, Delete duplicate objects, Bring or send Objects **Views:** Orthographic, Isometric, Cross sectional, Elevation, Auxiliary, Detail, Exploded views	Theory/Demo	4 HRS
6	Session 11&12	Exercises Included for practice	Practicals	4 HRS
7	Session 13	**LAYER MANAGEMENT** Layer Management Adding / Removing Layers Layer Status New Property Filter New Group Filter Layer Status Manager Plot Control **Object Selection Methods:** Select, Qselect, Filter	Theory/Demo	2 HRS
8	Session 14	Exercises Included for practice	Practicals	2 HRS
9	Session 15	**HATCH TECHNICS AND ISOMETRIC VIEW** **Hatching utilities:** Hatch, Hatchedit, Gradient, Boundary, Hatch Settings, Fill and Fill mode **Inquiry commands:** Id, Dist, List, Radius, Angle, Area, Volume **Isometric drawings:** Isometric snap and Isometric drawings	Theory/Demo	2 HRS
10	Session 16	Exercises Included for practice	Practicals	2 HRS
11	Session 17&18	**DIMENSIONING, TOLERANCING AND TABLE CREATION TECHNICS** **Dimensions:** Linear, Aligned, Radius, Diameter, Angle, Arc length, Continuous, Baseline, Tolerance, Dimension Space, Dimension Break, Inspection, Jogged radius, Ordinate dimensions, Leader, Qleader, Oblique, Dimensions Text angle, Dimensions Text Alignment, Quick dimension. **Leader:** Multi leader, Multi leader style, Add leader, Align Leader, Collect Leader, Remove Leader **Annotation Tools:** Text, Style, Multiline Text, Scale Text, Spell, Table, Table style, Table edit. Dimension style Manager, Dimension edit, Dimension text edit, Dimension Associative, Re-associate, Jog line, Inspect, Update	Theory/Demo	4 HRS
12	Session 19&20	Exercises Included for practice	Practicals	4 HRS
13	Session 21	**PARAMETRIC DRAWING** **Parametric Modeling:** Geometric Constraints, Dimensional Constraints, Delete constraints	Theory/Demo	2 HRS
14	Session 22	Exercises Included for practice	Practicals	2 HRS
		APPLICATION OF BLOCKS AND ATTRIBUTES		

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15	Session 23&24	Create Block, Base, Write block, Dynamic Block, Insert, Design Center, and Tool Palette Divide and Measure (Use block and point options) **Attributes:** Attribute definition, Attribute edit, Edit single & multiple attribute, Attribute Display, Manage Attributes, Attdia, Attext, Eattext Title blocks creation, Symbol creation, BOM creation, Data extraction and Link Data	Theoy/Demo	4 HRS
16	Session 25&26	Exercises Included for practice	Practicals	4 HRS
17	Session 27	**EXTERNAL REFERENCING AND EXPRESS TOOLS** **External References:** Xref, Xbind, Circular Xref, Refedit Action Recorder **Express Tools:** Explode Attribute, Arc Aligned, Text Explode, Text Case, Torient, Text Fit, Text circle and Modify text Super Hatch, Make Line type, Shape creation	Theory/Demo	2 HRS
18	Session 28	Exercises Included for practice	Practicals	2 HRS
19	Session 29	**OBJECT LINKING AND EMBEDDING** **OLE concepts:** Olelinks, Olescale, Insertobject, Hyperlink Copy, Copybase, Copylink, Pasteclip, Pastespecial, Pasteblock, Pasteoriginal	Theory/Demo	2 HRS
20	Session 30	Exercises Included for practice	Practicals	2 HRS
21	Session 31	**LAYOUT, PLOT AND PUBLISH** Etransmit, Publish, Publish to web Layout, Page setup Viewports, Viewport Lock and Unlock, Viewport Layers Management, Model space, Paper space Plot styles and Plot (Print) in Model and Layout area	Theory/Demo	2 HRS
22	Session 32	Exercises Included for practice	Practicals	2 HRS
23	Session 33 & 34	**INTRODUCTION TO AUTOCAD MECHANICAL** AutoCAD Mechanical Introduction Commands and Features Hole Features - through, tapped, counter bored and counter sunk, slots, and so on Fasteners - nuts, bolts, washers, screw connections, and so on. Shaft Parts - circlips, o-rings, seals, bearings, keys, and so on. Shaft Features - gears, tapers, grooves, keys, center holes, undercuts, and so on (Gears and tapers are found within the shaft generator) Steel Shapes - beams (I Shapes), angles (L Shapes), channels (U Shapes), Rectangular Hollow and Circular Hollow shapes, and so on Generators - shaft, spring, cam, chain and belt, and so on Fasteners and Shaft Couplings.	Theory/Demo	4 HRS
24	Session 35 & 36	Exercises Included for practice	Practicals	4 HRS
25	Session 37,38&39	**INTRODUCTION TO 3D MODELING IN AUTOCAD** AutoCAD 3D Introduction, Solid Modelling, Surface Modelling **Basic Modeling Tools:** Box, Cylinder, Cone, Sphere, Pyramid, Wedge, Torus, Extrude, Sweep, Revolve, Loft, Press pull Blend, Patch, Surface Offset, Surface Trim, Surface Fillet, Surface Extend, Surface untrim **Basic Editing Tools:** Union, Subtract, Intersect, Slice, Extrude Faces, Separate, Shell, Fillet, Chamfer, 3D Orbit, Move/Rotate/Scale Gizmo, Taper/extrude/Offset Faces UCS Co-ordinates, Viewport controls, View controls, Visual controls, Section, Import & Export options, Basic 3D Models **Layout:** Base View from Model space and Inventor, Projected, Section, and Detail views, Auto update and Update views	Theory/Demo	6 HRS
26	Session 40	Exercises included for Practice	Practicals	2 HRS